

Endpoint Failure for an Unbounded-Group Transference Principle

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Source and Target

The source paper is Markus Haase, *A Transference Principle for General Groups and Functional Calculus on UMD Spaces*, arXiv:0807.3906. In Section 3, Theorem 3.2 proves a transference estimate for a C_0 -group U satisfying

$$\|U(s)\| \leq M \cosh(\omega_0 s)$$

when the measure weight parameter ω is strictly larger than ω_0 . In Remark 3.3 the source says that this strict inequality is essential in the proof and that the endpoint $\omega = \omega_0$ is not expected.

This packet proves that the endpoint estimate indeed fails, already in the scalar Hilbert-space case.

Endpoint Form Refuted

At the endpoint, the analogue of Haase's Theorem 3.2 would assert the following kind of estimate: for $p = 2$, $X = \mathbb{C}$, a C_0 -group U with $\|U(s)\| \leq M \cosh(\omega_0 s)$, and all $\mu \in M_{\omega_0}(\mathbb{R})$,

$$\left| \int_{\mathbb{R}} U(s) \mu(ds) \right| \leq CM^2 \|L_{\mu_{\omega_0}}\|_{L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})}, \quad \mu_{\omega_0}(ds) = \cosh(\omega_0 s) \mu(ds).$$

Here L_{ν} denotes convolution by the finite measure ν .

Proof Intuition

Take the scalar group $U(s) = e^{\alpha s}$, whose growth type is exactly $\alpha > 0$. If $\nu = \mu_\alpha$, then the transferred scalar operator is pairing ν with

$$\frac{e^{\alpha s}}{\cosh(\alpha s)} = 1 + \tanh(\alpha s).$$

For scalar L^2 , the convolution norm of ν is just the supremum norm of its Fourier transform. Thus the endpoint estimate would say that pairing with $\tanh(\alpha s)$ is controlled by $\|\widehat{\nu}\|_\infty$, at least for measures of total mass zero.

This is false. The Fourier transform of $\tanh(\alpha s)$ has a principal value $1/\xi$ singularity at frequency 0. Smooth Fourier multipliers bounded by one and supported on $1/N \lesssim |\xi| \lesssim 1$ therefore pair with it like $\int_{1/N}^1 d\xi/\xi$, which grows as $\log N$.

Fourier Convention

We use

$$\widehat{f}(\xi) = \int_{\mathbb{R}} e^{-is\xi} f(s) ds, \quad f(s) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{is\xi} \widehat{f}(\xi) d\xi.$$

Lemma 1. *For $\alpha > 0$, in the sense of tempered principal-value distributions,*

$$\widehat{\tanh(\alpha \cdot)}(\xi) = \frac{\pi}{i\alpha \sinh(\pi\xi/(2\alpha))}.$$

Proof. The derivative of $\tanh(\alpha s)$ is $\alpha \operatorname{sech}^2(\alpha s)$. The standard contour computation gives

$$\mathcal{F}[\alpha \operatorname{sech}^2(\alpha \cdot)](\xi) = \frac{\pi\xi/\alpha}{\sinh(\pi\xi/(2\alpha))}.$$

Since differentiation corresponds to multiplication by $i\xi$, division by $i\xi$ gives the displayed formula away from zero. The distribution is odd and purely imaginary, so there is no additional delta mass at zero. $\square$

Theorem 1. *Let $\alpha > 0$. The endpoint transference estimate with $\omega = \omega_0 = \alpha$ fails for the scalar Hilbert space $X = \mathbb{C}$, $p = 2$, and $U(s) = e^{\alpha s}$.*

Proof. The group $U(s) = e^{\alpha s}$ satisfies

$$|U(s)| \leq 2 \cosh(\alpha s), \quad s \in \mathbb{R},$$

and its growth type is exactly α .

Assume, toward a contradiction, that an endpoint estimate holds with some finite constant C for this fixed group. Put

$$a = \frac{\pi}{2\alpha}.$$

Choose $\delta > 0$ so small that

$$|\sinh(a\xi)| \leq 2a|\xi| \quad (|\xi| \leq \delta).$$

For each sufficiently large N , let $\chi_N \in C_c^\infty(\mathbb{R})$ be even, $0 \leq \chi_N \leq 1$, with

$$\chi_N(\xi) = 1 \quad \text{for} \quad 2/N \leq |\xi| \leq \delta/2, \quad \text{supp } \chi_N \subset \{1/N \leq |\xi| \leq \delta\}.$$

Define

$$m_N(\xi) = -i \operatorname{sgn}(\xi) \chi_N(\xi).$$

Then $m_N \in C_c^\infty(\mathbb{R})$, $m_N(0) = 0$, and $\|m_N\|_\infty \leq 1$. Let

$$h_N(s) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{is\xi} m_N(\xi) d\xi, \quad \nu_N(ds) = h_N(s) ds.$$

Since m_N is smooth and compactly supported, h_N is a Schwartz function, hence ν_N is a finite measure. Its Fourier transform is $\widehat{\nu_N} = m_N$, and its total mass is zero because $m_N(0) = 0$. Therefore convolution by ν_N on scalar $L^2(\mathbb{R})$ has norm

$$\|L_{\nu_N}\|_{2 \rightarrow 2} = \|m_N\|_\infty \leq 1.$$

Now set

$$\mu_N(ds) = \operatorname{sech}(\alpha s) \nu_N(ds).$$

Then $\mu_N \in M_\alpha(\mathbb{R})$, since

$$\int_{\mathbb{R}} e^{\alpha|s|} |\mu_N|(ds) \leq 2\|\nu_N\| < \infty,$$

and $(\mu_N)_\alpha = \nu_N$. The transferred scalar operator is

$$\int_{\mathbb{R}} e^{\alpha s} \mu_N(ds) = \int_{\mathbb{R}} (1 + \tanh(\alpha s)) \nu_N(ds) = \int_{\mathbb{R}} \tanh(\alpha s) h_N(s) ds,$$

because $\nu_N(\mathbb{R}) = 0$.

By Lemma 1 and the definition of m_N ,

$$\begin{aligned}\int_{\mathbb{R}} \tanh(\alpha s) h_N(s) ds &= \frac{1}{2\pi} \left\langle \frac{\pi}{i\alpha \sinh(a\xi)}, m_N(-\xi) \right\rangle \\ &= \frac{1}{2\alpha} \int_{\mathbb{R}} \frac{\chi_N(\xi)}{|\sinh(a\xi)|} d\xi.\end{aligned}$$

For N large enough that $2/N < \delta/2$, the last integral is bounded below by

$$\frac{1}{2\alpha} \int_{2/N \leq |\xi| \leq \delta/2} \frac{d\xi}{2a|\xi|} = \frac{1}{2\alpha a} \log\left(\frac{\delta N}{4}\right),$$

which tends to infinity with N .

The hypothesized endpoint transference estimate would instead give

$$\left| \int_{\mathbb{R}} e^{\alpha s} \mu_N(ds) \right| \leq C 2^2 \|L_{\nu_N}\|_{2 \rightarrow 2} \leq 4C$$

for all N , a contradiction. Hence no endpoint estimate can hold. $\square$

Verification Notes

The construction is scalar and uses only Plancherel's theorem for the L^2 convolution norm. The measures ν_N are absolutely continuous finite measures because their Fourier transforms m_N are smooth and compactly supported. The logarithmic divergence comes solely from the principal-value $1/\xi$ singularity in the Fourier transform of $\tanh(\alpha s)$.

Novelty and Search Bounds

The run indexes were searched for arXiv:0807.3906, the source title, ‘endpoint transference’, ‘omega_0’, and close variants; no duplicate packet was found. A bounded web search on 2026-07-19 found later uses and interpolation variants of Haase’s unbounded-group transference principle, but no exact endpoint counterexample.

References

References

- [1] M. Haase, *A Transference Principle for General Groups and Functional Calculus on UMD Spaces*, arXiv:0807.3906.

- [2] E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*, Princeton Lectures in Analysis I, Princeton University Press, 2003.